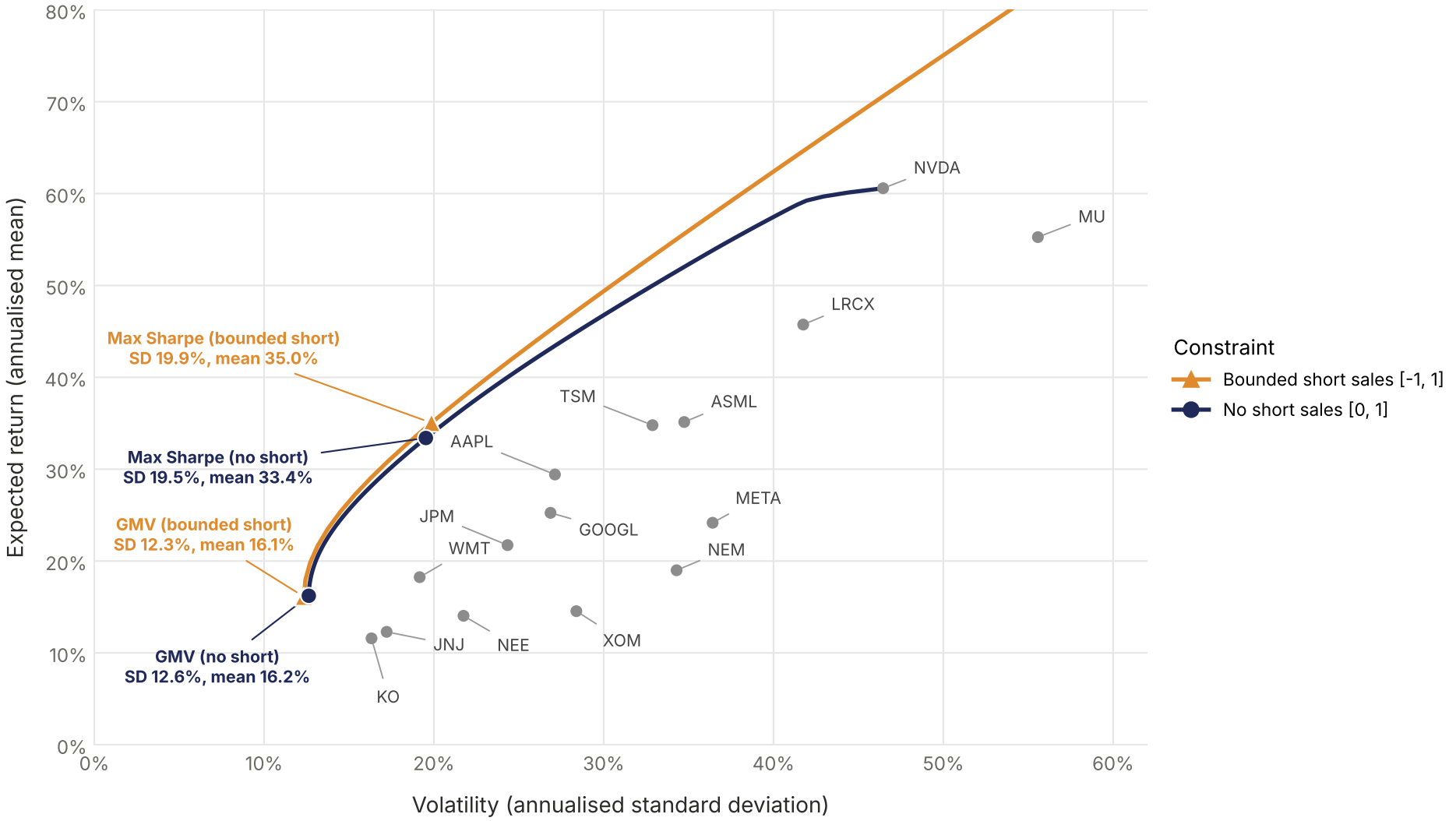
Figure 1: Efficient frontiers with and without short sales (monthly data, Oct 2016 to Sep 2026)



Data: Yahoo Finance via yfinance (adjusted for splits and dividends), downloaded 2026-11-04 11:32:29 UTC; USD. 120 monthly simple returns; mean x 12, SD x sqrt(12). No risk-free asset is held; r_f = 4.21% (3-month US T-bill, BEY, S&P Capital IQ, 3 Nov 2026) is used only for Sharpe ratios. The bounded-short frontier continues beyond the plotted area to 198.9% return at 207.8% SD.